

Gabriel Ganeles

gabriel.ganeles@gmail.com | gganeles.com | linkedin.com/in/gganeles

182 Neve America Technion Dormitories, Haifa IL, 3200003 | +972 58-712-0601

EDUCATION

Technion Israel Institute of Technology, Haifa, IL

Graduating April 2026

- Candidate for Bachelor's of Science in Biomedical Engineering, GPA: 84.9
- Relevant coursework by job description:
 - Software: Digital Signal Processing, Python, Deep Learning, Medical Instrumentation Design
 - Electrical: Electronic Circuit Theory, Semiconductors, Digital Signal Processing
 - Mechanical: Fatigue and Fracture Mechanics, Fluid Mechanics, CAD
 - Biomedical: Three Biomedical Lab Courses, Quantum Mechanics, MRI, Medical Instrumentation Design

SKILLS

3D Modeling: Creo, Onshape, SOLIDWORKS, Blender, Rhino, Grasshopper

EDA: LTSpice, Altium, KiCad

Programming Languages: Rust, Go, C, C++, Python, MATLAB, Javascript, HTML, CSS

Development Frameworks: Svelte, React, SolidJS, ROS2, Dioxus

Languages: English (Native), Hebrew (Fluent)

EXPERIENCE

Formula Technion / Haifa, IL

August 2023 - Present

Autonomous Team Co-lead

- Co-led a 12-member team to design and implement the autonomous systems for a Formula-style race car
- Engineered custom PCBs enabling high-speed encoder data acquisition, inter-CPU communication, and critical vehicle safety management
- Built and maintained the team's public website, increasing visibility and outreach to over 3,000 monthly visitors

Technion BioMotion Lab / Haifa, IL / Final Project for Degree

September 2024 - August 2025

Software Engineer

- Developed software to assist the Israeli Olympic windsurfing team in efficiently reviewing athlete footage
- Designed a graphical user interface (GUI) to enhance the accessibility of the algorithm
- Automated identification of maneuvers in windsurfing video footage using C++

Technion's Experimental Particle Physics Group / Haifa, IL

September 2023 - July 2024

Student Researcher

- Developed an algorithm to optimize detector overlap region selection for improved accuracy in Large Hadron Collider's muon detector
- Simulated muon interactions within ATLAS' New Small Wheel to study detector activation patterns

Flinker Lab at NYU Langone / New York City, NY

August 2023 - September 2023

Intern

- Extracted features from the sEEG data of 50 patients to train EEG-to-speech neural network (MATLAB)
- Convert medical images (MRI and CT) into 3D brain surfaces to map electrode placements into brain regions

Technion Rocketry Club / Haifa, IL

September 2022 - July 2023

Mechanical Engineer

- Created a spring-based 3D printed parachute launcher for a hybrid solid-gas model rocket
- Collaborated across propulsion and avionics teams to develop the rocket's structural body

Technion Automated Robotics Lab / Haifa, IL

January 2021 - July 2023

Robotics Engineer

- Built a 3D printed robotic gripper for reliable handling of salmon fillets
- Programmed an Arduino Nano in C++ to control motors and driver circuits

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INDEPENDENT PROJECTS

Gabe's Games Corner / set2.gganeles.com

September 2025

- Programmed 2 multiplayer games: Set and Anagrams (Snatch)
- Tech Stack: SolidJS, Tailwind, Rust+Warp

SET and Other Games / set.gganeles.com

January 2023

- Programmed 4 serverless multiplayer games in Svelte: Set, Anagrams (Snatch), Memory (Concentration), and Skull
- Tech Stack: Svelte, Firebase Hosting, Firebase Realtime Database, Tailwind-css

Shab-BOT / Whatsapp Bot

March 2022

- Developed in Go+Javascript, functionality includes: Shabbat meal guest tracking, location dependent Shabbat candle-lighting time, whatsapp-based reminders and scheduled messages.
- Tech Stack: Whatsmeow

AWARDS

Technion Bio-Hack Hackathon / First Place

April 2025

- Proposed a device to assist in the rapid prevention of cutaneous infection

Nucleate Bio-Fense Hackathon / First Place

June 2024

- One month bio-convergence hackathon sponsored by DDR&D (מפא"ד) and Teva
- Developed a solution to combat-based neck injuries

VOLUNTEERING

Gazelle Valley / Park Volunteer / JERUSALEM IL

September 2019 - March 2020

- Volunteered 3 hours weekly to care for the park, planting trees, cleaning litter, and clearing reeds from the swamp
- Constructed a cage for baby turtles to protect them from predatory birds